

Unit 12 Solutions

12.6 Design a circuit using D flip-flops that will generate the sequence 0, 0, 1, 0, 1, 1 and repeat. Do this by designing a counter for any sequence of states such that the first flip-flop takes on this sequence. There are many correct answers, but do not duplicate states, because each state can have only one next state.

0 → 0 → 1 → 0 → 1 → 1 6 states → 3 FFs

C	B	A	D_C^+	D_B^+	D_A^+
0	0	0	0	0	1
0	0	1	1	0	0
0	1	0	1	0	1
0	1	1	X	X	X
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	0	0	0
1	1	1	X	X	X

$D_C = A + B C'$
 $D_B = C B'$
 $D_A = C' A'$

BA	0	1
00	0	0
01	1	1
11	X	X
10	1	0

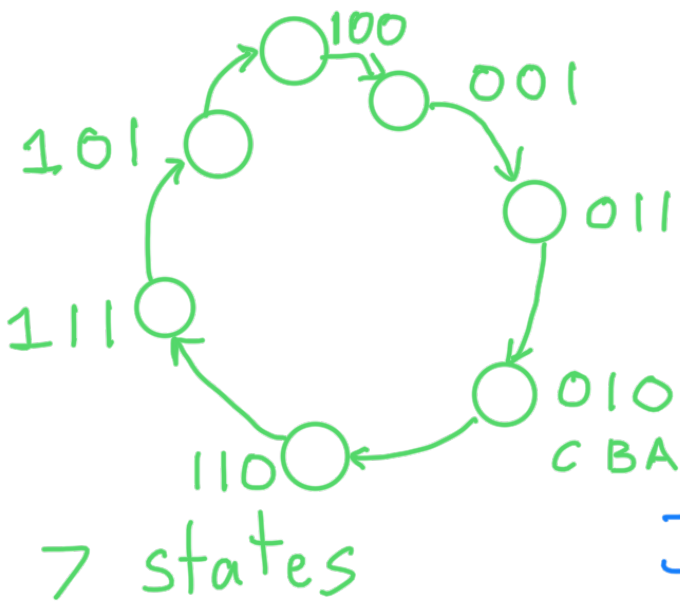
12.8 Design a 3-bit counter which counts in the sequence: 001, 011, 010, 110, 111, 101, 100, (repeat) 001, ...

(a) Use J-K flip-flops.
(b) Use S-R flip-flops.

In each case, what will happen if the counter is started in state 000?

J	K	Q	Q ⁺	Q	Q ⁺	J	K	Q	Q ⁺	J	K	Q	Q ⁺	J	K	Q	Q ⁺
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	0	0	1	1	0	1	0	1	0	1	0	1	0	1	0
0	1	1	0	0	1	1	1	0	1	1	1	0	1	1	1	0	1
1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	0	1	0
1	0	1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0
1	1	0	1	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	1	1	1	1	0	0	1	1	0	0	1	1	0	0	1	1

C	B	A	C ⁺	B ⁺	A ⁺	J _C	K _C	J _B	K _B	J _A	K _A
0	0	0	X	X	X	X	X				
0	0	1	0	1	1	0	X	1	X		
0	1	0	1	1	0	1	X				
0	1	1	0	1	0	0	X				
1	0	0	0	0	1	X	1				
1	0	1	1	0	0	X	0				
1	1	0	1	1	1	X	0				
1	1	1	1	0	1	X	0				



J _C	C	0	1
BA	00	X	X
01	0	X	
11	X		
10	1	X	

$J_C = A'$

K _C	C	0	1
BA	00	X	1
01	X	0	
11	X	0	
10	X	0	

$K_C = B'A'$

12.9 An M-N flip-flop works as follows:

If $MN = 00$, the next state of the flip-flop is 0.

If $MN = 01$, the next state of the flip-flop is the same as the present state.

If $MN = 10$, the next state of the flip-flop is the complement of the present state.

If $MN = 11$, the next state of the flip-flop is 1.

(a) Complete the following table (use don't-cares when possible):

Present State	Next State		
Q	Q^+	M	N
0	0	0	X
0	1	1	X
1	0	X	0
1	1	X	1

(b) Using this table and Karnaugh maps, derive and minimize the input equations for a counter composed of three M-N flip-flops which counts in the following sequence:

$CBA = 000, 001, 011, 111, 101, 100, (\text{repeat}) 000, \dots$

010, 110

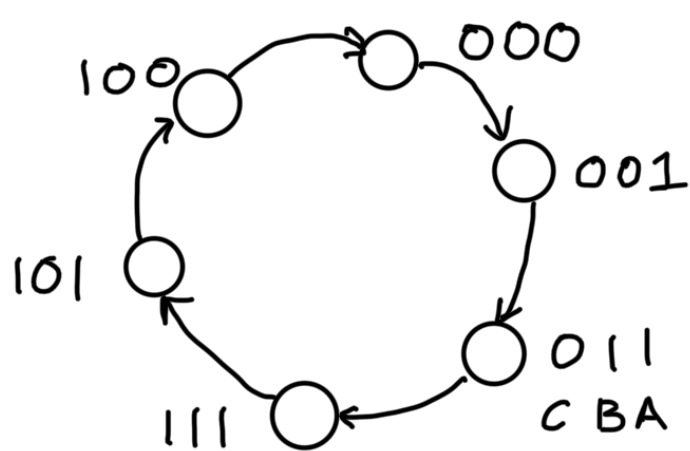
M	N	Q	Q ⁺
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

a)

Q	Q ⁺	MN
0	0	00 } 0X
0	1	10 } 1X
1	0	00 } X0
1	1	11 } X1

b)

C	B	A	C ⁺	B ⁺	A ⁺	M _C	N _C	M _B	N _B	M _A	N _A
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	1	0	X	1	X	X	1
0	1	0	X	X	X	X	X	X	X	X	X
0	1	1	1	1	1	1	X	1	1	X	1
1	0	0	0	0	0	X	0	0	X	0	X
1	0	1	1	0	0	X	1	0	X	X	0
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	1	0	1	X	1	X	0	X	1



M_C

	C	0	1
BA			
00	0	X	
01	0	X	
11	1	X	
10	X	X	

$$M_C = B$$

N_C

	C	0	1
BA			
00	X	0	
01	X		1
11	X		1
10	X		X

$$N_C = A$$

M_a

N_a

M_a

N_a

		C	
		0	1
B \ A	00	0	0
	01	1	0
	11	X	X
	10	X	X

$$M_B = C'A$$

		C	
		0	1
B \ A	00	X	X
	01	X	X
	11	1	0
	10	X	X

$$N_B = C'$$

		C	
		0	1
B \ A	00	1	0
	01	X	X
	11	X	X
	10	X	X

$$M_A = C'$$

		C	
		0	1
B \ A	00	X	X
	01	1	0
	11	1	1
	10	X	X

$$N_A = C' + B$$